

Multiple Choice (10 marks)

1. A grocery store sells a bag of 5 lemons for \$2.00. What is the unit cost of each lemon in the bag?
 - (a) \$2.50
 - (b) \$0.60
 - (c) \$0.40
 - (d) \$0.10
2. The mean of 7 numbers is 15. When an 8th number is added, the mean decreases to 12. What is the 8th number?
 - (a) -12
 - (b) -9
 - (c) 0
 - (d) 8
3. In parallelogram $ABCD$, angle B measures 110° . What is the number of degrees in the measure of angle C ?
 - (a) 145
 - (b) 90
 - (c) 60
 - (d) 70
4. If all the values of a data set are the same, all of the following must equal zero except for which one?
 - (a) Mean
 - (b) Standard deviation
 - (c) Variance
 - (d) Range
5. What expression can be used to show 270,240 written in expanded form?
 - (a) $200,000 + 7,000 + 200 + 4$
 - (b) $200,000 + 7,000 + 200 + 40$
 - (c) $200,000 + 70,000 + 200 + 40$
 - (d) $200,000 + 70,000 + 2,000 + 40$

True / False (10 marks)

Answer True or False

6. True or False: To pass the algebra test at Academic Academy, a student can miss at most 7 problems out of 35.
7. True or False: The probability that exactly two of four 20-sided dice show an even number is $\frac{1}{16}$.
8. True or False: The mean of the data set 18, 9, 9, 10, 11, 14, 30, 19 is 11.
9. True or False: The domain of the function $h(x) = (25-x^2) + (-(x-2))$ is an interval of width 5.
10. True or False: The value of $f(-2)$ for the function $f(x) = 8x^3 - 6x^2 - 4x + 5$ is 5.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the implications of a study that links green tea consumption to lower depression rates. What caution should be taken when interpreting such findings in health recommendations?
12. Discuss and analyze how the expressions $b + b + 3c$ and $2b + 3c$ are equivalent for any variable value greater than zero. Provide a detailed explanation of your reasoning.

End of Paper